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United States Patent	12403354
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Mosqueda; Aaron

Exercise device for strengthening gluteal muscles

Abstract

An exercise device for strengthening muscles is disclosed, having a continuous belt system that allows a user to move the continuous belt using their feet to exercise muscles of the person while the person lies in front of the continuous belt with their back braced on a resting surface. Such an exercise device allows a user to exercise their gluteus and hamstring muscles while strengthening their pelvic stabilizers. The exercise device could also have a weight device that allows a user to rest a weight on their torso during the exercise to increase the difficulty and effectiveness of the exercise.

Inventors:	Mosqueda; Aaron (Los Angeles, CA)
Applicant:	Mosqueda; Aaron (Los Angeles, CA)
Family ID:	90000641
Appl. No.:	18/505992
Filed:	November 09, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240066353 A1	Feb. 29, 2024

Related U.S. Application Data

continuation parent-doc US 17225046 20210407 US 11253743 20220202 child-doc US 17647951
continuation-in-part parent-doc US 17647951 20220113 US 11833387 child-doc US 18505992

Publication Classification

Int. Cl.: **A63B22/02** (20060101); **A63B21/00** (20060101); **A63B21/06** (20060101); **A63B21/065** (20060101); **A63B21/072** (20060101); **A63B21/078** (20060101)

U.S. Cl.:

CPC **A63B22/02** (20130101); **A63B21/0615** (20130101); **A63B21/065** (20130101); **A63B21/0724** (20130101); **A63B21/0783** (20151001); **A63B21/4039** (20151001); A63B2208/0252 (20130101)

Field of Classification Search

CPC: A63B (21/00181); A63B (21/06); A63B (21/0615); A63B (21/0616); A63B (21/0617); A63B (21/062); A63B (21/0622); A63B (21/065); A63B (21/072); A63B (21/08); A63B (21/4023); A63B (21/4025); A63B (23/02); A63B (23/0205); A63B (23/0216); A63B (23/0233); A63B (23/04); A63B (23/0405); A63B (23/0411)

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Primary Examiner: Moore; Zachary T

Attorney, Agent or Firm: DANE IP Law PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation in part application of U.S. Ser. No. 17/647,951, filed on 2022 Jan. 13, which is a continuation application of U.S. patent application Ser. No. 17/225,046, filed on 2021 Apr. 7, the entire contents of which is expressly incorporated herein by reference.

FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not Applicable

BACKGROUND

(2) Exercise devices, such as bench press racks, treadmills, and leg curl machines, are commonly purchased and used by consumers in order to strengthen their muscles at home or in a gym. There are few machines that could be used, however, to exercise gluteal muscles.

(3) It would therefore be desirable to provide improved exercise devices and methods of using exercise machines to strengthen a person's gluteal muscles during a workout.

SUMMARY OF THE INVENTION

(4) A system and method for exercising gluteus and hamstring muscles of a person are disclosed. The method involves disposing an exercise machine base having a conveyor belt or a continuous belt under the feet of a person while the person lies back-down on a resting surface. In some embodiments, the continuous belt could be pulled taught between at least two rollers. The continuous belt preferably is configured to move about the at least two rollers such that the continuous belt rolls around the at least two rollers. The person braces their upper back against a friction surface to hold the upper back of the person in place relative to the exercise machine while the person moves the continuous belt using their feet. By moving the continuous belt with the person's feet, the person can effectively exercise their buttock muscles while targeting their pelvic stabilizers. For example the person may use the balls of their feet to push the continuous belt away from themselves or pull the continuous belt towards themselves.

(5) A person could increase the difficulty of the exercise by raising their torso while maintaining a minimum distance between a resting surface and the buttock muscles of the person while performing the exercise. Such a resting surface could be, for example, a resting surface that supports the base of the exercise machine, a back brace that is coupled to the exercise machine base, or a yoga mat or carpet that sits between the ground and the base of the exercise machine. The resting surface typically acts as the friction surface that the person uses to brace their upper back in place while the person moves the belt of the exercise machine.

(6) The difficulty of the exercise can be further increased by resting a weight on the person's torso while performing the exercise. Such a weight could be positioned in any suitable location, such as the pelvis area, chest area, or the ab area of the person performing the exercise. In some embodiments the weight could be coupled to a weight bar. The weight bar could have one or more weight posts that hold the weight in place during use, such as at the center of the weight bar or on the sides of the weight bar, oriented in any suitable direction. The weight bar could be attached to the exercise machine, such as the base of the exercise machine, and could be pivotably attached to the exercise machine to allow the weight bar to move with the person as the person raises and lowers their torso.

(7) Any machine having a continuous belt that can be moved with a person's feet could be used to perform this exercise. In some embodiments, an exercise machine could be provided that has a base and a weight bar. The base could have a continuous belt pulled taught between at least two rollers, configured to move along the rollers when a person's feet applies a force on the surface of the continuous belt. The rollers could have one or more mechanisms that increase or decrease a resistance of the continuous belt as it moves about the rollers. Such a mechanism could comprise a brake mechanism that alters a friction force applied to a roller, or a torque multiplier that changes a torque force applied to an axle with differing settings. The base could have a relatively thin thickness, which allows a person lying on a resting surface to easily place their feet on the top surface of the continuous belt. In such embodiments, the top surface of the belt could be positioned less than 3, 2, 1, or $\frac{1}{2}$ feet from the top of a resting surface. The top surface of the continuous belt could be made of any suitable material having a coefficient of friction greater than 1 with a person's feet, socks, or shoes, such as rubber, plastic, or textured cloth.

(8) The weight bar could be coupled to the base comprising the continuous belt, and could have at least one weight attachment device that holds one or more weights in place relative to the weight bar. The weight attachment device is preferably configured to evenly distribute weight on the user's torso. For example, the weight attachment device could comprise a weight post disposed at a center of the weight bar or could comprise a plurality of weight posts disposed opposite each other, equidistant from the center of the weight bar. The weight post could comprise a cylindrical protrusion having a diameter of at most 1, 2, or 3 inches. In some embodiments, a clip could be removably attached to the weight post to hold the weight in place while a user exercises using the machine. Such clips could be frictionally coupled to the weight post, for example by using a spring that releases from the weight post when force is applied to the spring and compresses against the weight post when force is not applied to the spring, or by using a clamp that could be tightened or loosened by turning a threaded fastener.

(9) The weight bar is preferably disposed to rest on a torso of a user of the exercise machine while the user applies a pushing or pulling force to the continuous belt with their feet. The weight bar could comprise any material suitable for holding a weight, for example a metal, such as aluminum or steel, or a hard plastic. In some embodiments, the weight bar comprises a tubular or rectangular material that is bent or cut into shape.

(10) The weight bar could have a pad or a cushion wrapped around a section of the weight bar that protects the user's torso from the force of the hard weight bar. In some embodiments, the weight bar could be removably coupled to the base in any suitable manner, for example the weight bar could be hingedly connected to the base with a threaded fastener to pivot while the user raises and lowers the weight bar, and the threaded fastener could be loosened to detach the weight bar from the base. In some embodiments, the weight bar could be configured to rotate between at least 90, 120, or 150 degrees at a pivot point. The weight bar could comprise a bracket rotateably attached to a frame of the belt system, where the bracket has a cross-bar that supports the weight on the user's stomach. Such a bracket could have one, two, or more extension members that are equal to one another in length, which are coupled to the base near a proximal end. The extension members are preferably configured such that the cross bar is positioned above a user's stomach or hips and/or below the

user's shoulders during use of the exercise device. For example, the extension members could have a length between 2.5 feet and 3.5 feet. In some embodiments, the extension members could be configured to be extendable and/or retractable, for example by using telescoping extension members. In some embodiments, the weight bar could alternatively, or could also, be coupled to the base via a sliding mechanism that allows the coupling location of the weight bar to shift its position along an axis of the base. Preferably, the coupling location of the weight bar could shift its position along the major length of the base along the length of a major frame member.

(11) The weight bar could have a flattened or rounded portion alongside that allows the weight bar to comfortably rest on the torso of the user. In some embodiments, the weight bar could comprise a substantially circular, rectangular, or square cross-sectional area to form a round bar or rectangular tube. The weight bar could also have one or more supports that holds the weight bar above a torso of the user when not in use. Such a support could be hingedly coupled to the weight bar in the similar manner as a bicycle kickstand to rotate between a support position that holds the weight bar above the torso of the user and a standby position that allows the weight bar to rest on the torso of the user. The support could be configured to pivot by at least 50, 80, 90, or 120 degrees.

(12) Furthermore, an exercise device for strengthening gluteus muscles and hamstrings is disclosed. The exercise device may have a base having a belt system with a continuous belt that is taut between at least two rollers, a back support attached to the front of the base, the back support having a lying surface between an outer frame of the back support, the outer frame having a first longitudinal side and a second longitudinal side, the first longitudinal side having a first lengthwise slot and the second longitudinal side having a second lengthwise slot, a first attachment component slidably coupled to the first lengthwise slot of the first longitudinal side of the outer frame, and a second attachment component slidably coupled to the second lengthwise slot of the second longitudinal side of the outer frame. The first and second attachment components may be configured to have a resistance band attached thereto for the resistance band to apply a downward force on a torso of a user when the user is lying on the lying surface of the back support.

(13) In some embodiments, the exercise device may have a first gripping handle slidably coupled to a third lengthwise slot of the first longitudinal side, and a second gripping handle slidably coupled to a fourth lengthwise slot of the second longitudinal side. In some embodiments, the first lengthwise slot may be located on a first inner surface of the first longitudinal side and the second lengthwise slot may be located on a second inner surface of the second longitudinal side. In some embodiments, the third lengthwise slot may be located on a first upper surface of the first longitudinal side and the fourth lengthwise slot may be located on a second upper surface of the second longitudinal side.

(14) In some embodiments, the first and second longitudinal sides may be made from extrusion profile frames. In some embodiments, the first and second lengthwise slots may be either V-slots or T-slots.

(15) In some embodiments, the first attachment component may have a first body having a first attachment ring on one end of the first body and a first sliding shaft on another end of the first body, the first sliding shaft inserted through an attachment block of the first attachment component. In some embodiments, the first gripping handle may have a shaft body with a gripping portion on one end of the shaft body and a sliding shaft on another end of the shaft body, the sliding shaft inserted through an handle block of the first gripping handle.

(16) In some embodiments, the first and second attachment components each may have an attachment ring. In some embodiments, the first and second attachment components each may have a snap hook.

(17) Additionally, an exercise device for strengthening gluteus muscles and hamstrings is disclosed. The exercise device may have a base having a belt system with a continuous belt that is taut between at least two rollers, the belt system being at an incline, a back support attached to the front of the base, the back support having a lying surface between an outer frame of the back support, the

outer frame having a first longitudinal side and a second longitudinal side, the first longitudinal side having a first lengthwise slot and the second longitudinal side having a second lengthwise slot, a first attachment component slidably coupled to the first lengthwise slot of the first longitudinal side of the outer frame, and a second attachment component slidably coupled to the second lengthwise slot of the second longitudinal side of the outer frame. The first and second attachment components may be configured to have a resistance band attached thereto for the resistance band to apply a downward force on a torso of a user when the user is lying on the lying surface of the back support. The base may be elevated height-wise relative to the back support and the belt system inclines downwards towards the back support.

(18) In some embodiments, the exercise device may have a first gripping handle slidably coupled to a third lengthwise slot of the first longitudinal side, and a second gripping handle slidably coupled to a fourth lengthwise slot of the second longitudinal side. In some embodiments, the first lengthwise slot may be located on a first inner surface of the first longitudinal side and the second lengthwise slot may be located on a second inner surface of the second longitudinal side. In some embodiments, the third lengthwise slot may be located on a first upper surface of the first longitudinal side and the fourth lengthwise slot may be located on a second upper surface of the second longitudinal side.

(19) In some embodiments, the first and second longitudinal sides may be made from extrusion profile frames. In some embodiments, the first and second lengthwise slots may be either V-slots or T-slots.

(20) In some embodiments, the first attachment component may have a first body having a first attachment ring on one end of the first body and a first sliding shaft on another end of the first body, the first sliding shaft inserted through an attachment block of the first attachment component. In some embodiments, the first gripping handle may have a shaft body with a gripping portion on one end of the shaft body and a sliding shaft on another end of the shaft body, the sliding shaft inserted through an handle block of the first gripping handle.

(21) In some embodiments, the first and second attachment components each may have an attachment ring. In some embodiments, the first and second attachment components each may have a snap hook.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a front perspective view of an exercise device in a storage position.

(2) FIG. 2 shows a front perspective view of the exercise device of FIG. 1 being unfolded to an in-use position.

(3) FIG. 3 shows a front perspective view the exercise device of FIG. 1 unfolded to an in-use position, with a human user using the machine to work out.

(4) FIG. 4 shows a side plan view of the exercise device of FIG. 3.

(5) FIG. 5 shows a front perspective view of an alternative exercise device.

(6) FIG. 6 shows a front perspective view of another exercise device.

(7) FIG. 7 shows a front perspective view of yet another exercise device.

(8) FIG. 8A shows a front perspective view of yet another exercise device.

(9) FIG. 8B shows a front perspective view of the exercise device of FIG. 8A with the attachment rings and gripping handles adjusted.

(10) FIG. 8C shows a front perspective view of the exercise device of FIG. 8B with a user using the device with a resistance band attached to the attachment rings.

(11) FIG. 8D shows a cross-sectional view of the back support of the exercise device of FIG. 8A.

(12) FIG. 9A shows a perspective view of the attachment ring of the exercise device of FIG. 8A.

(13) FIG. 9B shows a perspective view of the gripping handle of the exercise device of FIG. 8A.
(14) FIG. 9C shows a front view of the resistance band used with the exercise device in FIG. 8C.
DETAILED DESCRIPTION

(15) The following detailed description describes a novel exercise system and method to exercise the buttock muscles of a person.

(16) Various exercise devices could be utilized to perform such exercises. Four different exemplary devices are shown in FIGS. 1-7 as devices **100**, **100a**, **100b**, and **100c**, as examples. An exemplary device typically has a continuous belt system that can be placed in front of a user's feet during the exercise. Contemplated continuous belt systems include a base having a continuous belt mounted to a frame that is pulled taught between at least two rollers. The belt could be configured to rotate about the rollers, having an upper side of the belt that is positioned less than two feet from a resting surface that the base rests upon, such as the ground. The base comprising the continuous belt could have one or more sets of wheels coupled to a side of the base that allow a user to easily wheel the base around before and after a workout session.

(17) A weight bar could also be provided that is configured to rest on the person's torso during the workout. When the person lifts their pelvis off of the ground during the workout, the added weight from the weight bar provides added resistance during the workout. The weight bar could be configured to hold one or more weights on suitable weight attachment mechanisms, such as weight posts oriented in any suitable direction. One or more clips could be attached to such weight posts to hold the weights in place. The weight bar could be pivotally coupled to the base comprising the continuous belt. The weight bar could also have a kickstand-like support that can hold the weight bar above the person's torso to rest in between exercise sets. Also, a resistance band may be used in lieu of or in addition to the weight bar.

(18) The exercise is conducted by a person who lies in front of a continuous belt with their back braced against a friction surface near the continuous belt and their feet positioned over the continuous belt. The person may push the continuous belt away from themselves or pull the continuous belt towards themselves using their feet to exercise and strengthen their gluteal muscles and pelvic stabilizers during a workout. The difficulty of the exercise can be increased by increasing a tension of the continuous belt to make it more difficult to move with the feet, or by raising the pelvis of the person off of the ground during the workout. A weight bar may be placed on the torso of the user, such as the chest, abs, or pelvis of the person, to further increase the difficulty of the workout. In embodiments where the weight bar is pivotally coupled to the base comprising the continuous belt, the person could conduct the exercise by placing weights on the weight bar before a workout, pivoting the weight bar off of the ground, crawling under the weight bar to rest it on the person's chest, lifting their torso, and then moving the continuous belt with their feet while maintaining holding their pelvis off of the ground.

(19) FIG. 1 shows an exemplary exercise machine **100** in a storage configuration. The exercise machine **100** has a base **110**, a pair of handles **120**, and a weight bar **130**. The base **110**, the handles **120**, and the weight bar **130** are configured to be rotatable relative to one another to allow the three elements to fold into substantially parallel positions relative to one another to allow for easy storage in a vertical arrangement. Here, the pair of handles **120** are configured to be stationary as the base **110** is rotated relative to the pair of handles **120**, and the weight bar **130** is rotated relative to the base **110**. As used herein, elements that are “substantially parallel” to one another have a major axis along the greatest length of the element that are parallel within 10 degrees from one another. For example, the weight bar **130** is shown as having a major axis that is about 3 degrees offset from the major axis of the base **110**.

(20) The handles **120** comprise a stabilizer **125** which is coupled to the left handle **121** via the bracket **123** and is coupled to the right handle **122** via the bracket **125**. While the handles **120** are shown as separate components coupled together via brackets having threaded connector, the handles **120** could be coupled to the stabilizer **125** in any other suitable manner, or could be molded

into a single unitary element having the handles and stabilizer. The left handle **121** is rotatably coupled to the base **110** via a threaded fastener **126** that threads through holes (not shown) in both the left handle **121** and the base **110**, while the right handle **122** is rotatably coupled to the base **110** via a threaded fastener **127** that threads through holes (not shown) in both the right handle **122** and the base **110**. This allows the base **110** to rotate relative to the handles **120** via the pivot points located at the threaded fasteners **126** and **127**, respectively. Each of the handles, the left handle **121** and the right handle **122** have handle cushions **123**, which act to cushion any force applied to the hands of a user that grabs the cushions, similar to cushion **133**. In some embodiments, the handle cushion **133** is made from a material that is stiffer than the cushions that are coupled to the left handle **121** and the right handle **122**. The rotational position of the base **110** relative to the handles **120** could be locked or unlocked in place by tightening and untightening the threaded fasteners **126** and **127**, respectively, as needed. While the base **110** and the handles **120** are shown as rotatably coupled to one another via threaded fasteners, other rotatable fastening mechanisms could be used, such as hinges, nuts and bolts, ball joints, axles.

(21) In contrast, the weight bar **130** comprises a body **132** coupled to the base **110** via the fastening mechanism **136**, shown here as a nut, bolt, and washer, although other rotatable fastening mechanisms could be used, as disclosed herein. It should be understood that, on the other side of the base **110** along the length of the left support **104**, another fastening mechanism could be coupled to the base **110**, opposite to the fastening mechanism **136**, to provide two hinges upon which the weight bar **130** rotates.

(22) In some embodiments, the body **132** could have a length that is adjustable. For example, as shown in FIG. 2, the body **132** has a telescoping member **132b** configured to slide into a cavity (not shown) of the telescoping member **132a**. The telescoping member **132b** is coupled to the telescoping member **132a** using the tightening mechanism **132c**, which locks the positions of the first and second telescoping members **132a**, **132b** in place relative to one another when tightened, and which allows the first and second telescoping members **132a**, **132b** to slide relative to one another when loosened. Such a tightening mechanism could be configured in any suitable manner to lock and unlock the sliding members in place relative to one another. Other mechanisms could be used to fix the position of telescoping members relative to one another, such as a peg with a spring that applies pressure to the peg to allow the peg to extend into a hole. A latch could be used to compress the spring until the peg is aligned with a hole that a user wishes the peg to extend into. As explained in further detail below, in other embodiments, the extension members of the body **132** could have a static length, and the position of the weight bar **130** on a user's torso **210** could be adjusted instead or in addition to by sliding the attachment mechanism **136** along a length of the base **110**.

(23) In some embodiments the fastening mechanism **136** could slide along a major axis of the length of the right support **103**, such as in the embodiment shown in FIG. 7, as explained below, to allow the pivot point of the weight bar **130** to change along the length of the base **110**. It should be understood that, in embodiments where the opposite side of the weight bar **130** has a fastening mechanism coupled to the left support **104**, such a fastening mechanism would also slide along the length of the left support **104** along a similar rail. In such an embodiment, the weight bar **130** could be adjusted relative to the base **110** to rest on various sections of the user's torso **210**. In addition, allowing the fastening point of the weight bar **130** to slide relative to the length of the base **110** could allow the weight bar **130** to slide forward towards the rollers **102** such that the weight bar **130** rests just below the rollers **105**. Such a configuration allows the exercise machine **100** to be used as a standing treadmill-like leg exercise device rapidly without needing to fold the weight bar **130** underneath the base **110** or removing the fastening mechanism **136**.

(24) Since the fastening mechanism **136** comprises a washer, the entire body **132** freely rotates relative to the base **110** and does not lock the relative rotational position of the base **110** relative to the weight bar **130**. This allows the weight bar **130** to freely move during the exercise without a

user needing to tighten or loosen the fastening mechanism **136**. By rotatably coupling the base **110**, the pair of handles **120**, and the weight bar **130** to one another, a user could fold the exercise machine **100** between a storage configuration shown in FIG. **1** and a ready-to-use position shown in FIG. **3** via the intermediary position shown in FIG. **2**, as is explained in more detail below.

(25) In some embodiments, the tighteners **126** and **127** could also be coupled to the roller **102**, allowing the tighteners **126** and **127** to also act as a tension brake for the continuous belt **105**. As the tighteners **126** and **127** are tightened, the friction force of the roller **102** could be increased, providing additional resistance for the continuous belt **105**, and allowing the tighteners **126** and **127** to provide the dual functionality of providing a tension brake for the continuous belt **105** as well as a stabilizing force to prevent the base **110** from rotating relative to the handles **120** when the exercise machine **100** is in the storage configuration shown in FIG. **1**.

(26) The base **110** comprises a frame having a right support **103**, a left support **104**, and a continuous belt **105** that is pulled taught between two rollers **101** and **102**. While two rollers are shown, more or less rollers could be used in other embodiments. The base **110** also comprises a platform **106** coupled to supports **107** between the left support **104** and the right support **103**, which provides a stabilizing force under the continuous belt **105** when a user places their feet on continuous belt **105** and applies a force to the continuous belt **105** so that the continuous belt **105** rotates on the rollers **101** and **102**. The base **110** also comprises feet **109** placed at the corners of the base **110** to help distance the continuous belt **105** from a resting surface, such as the ground, when the base **110** is disposed in the ready-to-use position. In some embodiments, the base **110** could be completely detached from the handles **120** to allow for a user to use the base **110** by itself as an exercise device. Such embodiments could be advantageous in areas with limited space.

(27) The weight bar **130** comprises a body **132** rotatably coupled to the base **110** via the fasteners **136**. The weight bar **130** has a weight post **131** that projects from a center of the weight bar **130**, which is configured to hold one or more weights **135** when in use. A cushion member **133** is wrapped around the weight bar **130** to help distribute and soften the force from the weight bar **130**. In some embodiments, the cushion member **133** comprises a softer material than the material that comprises the body **132**. For example, the body **132** could comprise a thermoplastic material or a metal material, such as aluminum, and the cushion member **133** could comprise a softer material such as a foam material or a cloth padding.

(28) While the weight bar **130** is shown in FIG. **4** as being a bent tube having a substantially circular cross-sectional area that rests on the torso **210** of the user **200**, the weight bar **130** could comprise any suitable cross-sectional area, for example an ovoid, rectangular or a square cross-sectional area. Such a weight bar that has a lower surface that rests on the torso **210** of the user **200** which is flatter would rest easier on the torso **210** of the user **200** during use.

(29) As shown in FIGS. **3** and **4**, in the ready-to-use position, a user **200** could place one or more weights on the weight bar **130**, lift the weight bar **130**, and dispose a portion of their body under the weight bar **130** to position the cushion material **133** on a portion of the torso **210** of the user **200**. Such a position allows the user **200** to exercise their back and gluteal muscles by raising the weight bar **130** a distance **302**, such as 1 in., 6 in., or 12 in. The user **200** could also place one or both of their feet **230** on the surface of the continuous belt **105**, allowing the user **200** to exercise their gluteal muscles by moving the continuous belt **105** with the ball of their feet **230** in either direction.

(30) In one embodiment, a user **200** uses the exercise device **100** by lying down in front of the exercise device **100** with their feet on the continuous belt **105**. The user **200** braces their back **240** against a surface, such as resting surface **300**, such that the user remains in place relative to the exercise device **100**, even while moving their feet against the continuous belt **105**. Contemplated resting surfaces include a floor of a room, a carpet, a rug, or an exercise mat. Such resting surfaces **300** preferably have a coefficient of friction greater than 1 with materials covering the back **240**, such as cloth or human skin. Once the user **200** is sufficiently braced, the user **200** could then move the top surface of the continuous belt **105** in either direction using their feet **230**, such as moving

the top surface of the continuous belt **105** towards the roller **102** or towards the roller **101**.

(31) FIG. 5 shows an alternative exercise device **100a** having an alternate weight bar **130a**. The weight bar **130a** has two weight posts **131a** and **133a** disposed on either side of the weight bar **130a**. The weight **132a** is coupled to the weight post **131a** while the weight **134a** is coupled to the weight post **133a**. The weight posts **131a** and **133a** are preferably disposed an equal distance from the center of the body of the weight bar **130a**, which balances the weight bar **130a** on user **200** when the weights **132a** and **133a** are of equal mass.

(32) The weight bar **130a** also has supports **141** and **142** that are rotatably coupled to the weight bar **130a** via the attachment mechanisms **144**. The attachment mechanisms **144** are shown here as nuts, bolts, and washers, but could be any suitable rotatable attachment mechanism, as discussed herein. In some embodiments, the attachment mechanisms **144** could comprise a kickstand that is biased to maintain stability in an engaged position, as shown, or a disengaged position that is substantially parallel to the side bars of the weight bar **130a**. The supports **141** and **142** act to hold the weight bar **130a** above the torso **210** of the user **200** when the user **200** wants to take a rest, or wants to perform an exercise without needing to move the weight bar **130a** to an alternative position out of the way. The supports **141** and **142** could be configured to pivot by at least 50, 80, 90, or 120 degrees from an origin angle parallel to the body **132** of the weight bar **130a**. In some embodiments, either of the supports **141** and **142** could have a fastening system (not shown) similar to any of the fastening systems disclosed herein to fix the angle of the support in place relative to the body **132** of the weight bar **130a**. In other embodiments, a support could be configured to freely rotate about an axis to allow the supports **141** and **142** to hang in accordance to a gravitational force.

(33) The supports **141** and **142** could be configured to have a plurality of lengths in some embodiments. For example, the exercise device **100a** could be configured to have a plurality of supports that could be coupled to the weight bar **130a**, where pairs of supports are of differing lengths of other pairs of supports. For example, one pair of supports could have a length of 8 inches while another pair of supports could have a length of 12 inches. In other embodiments, the supports **141** and **142** could be comprised to have an adjustable length, for example by comprising a pair of telescoping or sliding elements relative to one another, and a locking mechanism that locks one telescoping or sliding element relative to the other, in a similar manner to the telescoping elements **132a** and **132b** shown in FIG. 2. By providing supports having adjustable lengths, the weight bar **130a** could be configured to rest at different heights while in a standby position, to accommodate users having differing chest thicknesses.

(34) FIG. 6 shows another exercise device **100b** having a back support **150** that is used to provide a friction surface **151** that the user **200** could use to brace themselves against relative to the base **110**. The back support **150** is coupled to the base **110** via attachment mechanism **136**, which allows the back support **150** to rotate relative to the base **110** while still ensuring that the back **240** of the user **200** maintains a minimum threshold distance from the base **110** while exercising. While the back support **150** is shown as substantially flat to allow for users of many different heights to use the back support **150**, in some embodiments back support **150** could have a curved back to match a height of a user. The top surface of back support **150** preferably comprises a surface having a high friction force when touching skin or clothing, such as rubber or plastic, which has a coefficient of friction greater than 1 with bare skin or clothing. Providing such a back support **150** with the exercise device **100b** allows a user to use the exercise device **100b** on any suitable surface that has a low friction force, such as a slippery wooden floor in a gym or a studio. In some embodiments, the back support **150** could be configured to have an adjustable length mechanism that allows for users of different sizes to use the same back support **150**. For example, one or both of the bars along the side of the back support **150** could telescope in a similar manner to the body **132** of the weight bar **130**, allowing the position of the top bulging edge of the back support **150** to move its distance from the base **110**.

(35) FIG. 6 also shows a weight clip **170** which is used to hold the weight **135** in place on the weight post **131** during use. The weight clip **170** is shown as a spring that is biased to tighten when a user **200** does not apply a clamping force to the weight clip **170**, and to loosen when a user **200** applies a clamping force to the weight clip **170**, but could be any suitable clip that is used to hold a weight **135** in place on a weight post **131** during use, such as a clamping clip with a threaded fastener that tightens and loosens the clip as the threaded fastener is rotated in either direction. A user **200** could remove the weight clip **170** from the weight post **131** when the user **200** wishes to remove or place weights **135** on the weight post **131**, and could then couple the weight clip **170** to the weight post **131** to hold the weights **135** in place on the weight post **131**.

(36) FIG. 7 shows yet another exercise device **100c** having an alternative weight bar **130c** having the weight posts **136c** and **138c** that extend outwardly from the center of the weight bar **130c**. The weight **137c** is shown coupled to the weight post **136c** while the weight **139c** is coupled to the weight post **138c**, although more or less weights could be coupled to the weight posts **136c** and **138c** in other embodiments. The weight posts **136c** and **138c** are preferably disposed an equal distance from the center of the body of the weight bar **130c**, which balances the weight bar **130c** on user **200** when the weights **137c** and **139c** are of equal mass. In some embodiments, the exercise device **100c** could have one or more weight clips, such as the weight clip **170** that holds the weights **137c** and **139c** in place on the weight posts **136c** and **138c** during use. While the weight posts **136c** and **138c** are shown here as 4-inch projections from the sides of the body of the weight bar **130c**, the weight posts could be any suitable length, such as projecting at most 3 inches or projecting at least 5 inches, in other embodiments.

(37) The weight bar **130c** also has a padding **133c** that is fastened to the weight bar **130c** using the fasteners **132c**. The padding **133c** extends from a bottom of the weight bar **130c** to rest on the torso **210** of the user **200**, allowing the compressive force from the weight bar **130c** to distribute along the lower surface of the padding **133c**. In some embodiments, alternative paddings (not shown) could be provided that could be interchangeable coupled to the weight bar **130c** to provide for multiple different weight distribution configurations to be used on the torso **210** of the user **200**.

(38) The fastening mechanism **136** of the weight bar **130c** is also coupled to a sliding rail **137**. Generally, the opposing side (not shown) of the exercise device **100c** would also have a similar sliding rail, which would allow the weight bar **130c** to slide along the length of the right support **103**. Preferably, when a user tightens the fastening mechanism **136**, the fastening mechanism **136** would fix the weight bar **130c** in place along the length of the right support **103** to prevent the weight bar **130c** from sliding along the length of the right support **103**, while still allowing the weight bar **130c** to pivot relative to the base **110**. Such functionality could be achieved by using any suitable means, for example by disposing a compressive washer between the weight bar **130c** and the base **110** and a non-compressive washer between the weight bar **130c** and the wing nut. Such a configuration allows the user **200** to adjust the position of the weight bar **130c** along the length of the user's torso **210**, such that the weight bar **130c** could be configured to comfortably rest on the user's stomach, solar plexus, or chest. Such a configuration also allows users of different heights to use the same weight bar in the same manner.

(39) In some embodiments, the sliding rail **137** could comprise a separate locking mechanism to lock the weight bar **130c** in place along the length of the sliding rail **137**, in addition to, or separate from the fastening mechanism **136**. For example, at the end of the sliding rail **137**, a spring-loaded ball detent **137a** could be provided that locks the end of the weight bar **130c** in place when the fastening mechanism **136** of the weight bar **130c** slides to interact with the spring-loaded ball detent **137a**. A user **200** could loosen the fastening system **136** to slide the weight bar **130c** along the sliding rail **137**, slide the weight bar **130c** such that the end of the weight bar **130c** locks with the spring-loaded ball detent **137a** while resting the handle of the weight bar **130c** underneath the base **110** next to the feet **109**, and then could use the base **110** without the weight bar **130c**. When the user **200** wishes to use the weight bar **130c**, the user **200** could push the spring-loaded ball

detent **137a** to unlock the weight bar **130c** to slide it closer to the roller **101**, and then could lock the weight bar **130c** in place using the fastening mechanism **136**. Any suitable locking mechanism that prevents the weight bar **130c** from sliding along the length of the sliding rail **137** could be utilized, such as an orthogonal path downwards that the fastening mechanism **136** falls into while the weight bar **130c** is slid forward, and must be pulled out of in order for the weight bar **130c** to slide back to the ready in-use position shown in FIG. 7.

(40) In another embodiment, a portion of the weight bar **130c** could comprise a mechanism separate from the fastening mechanism **136** that mates with a portion of the base **110** to lock in place. For example, one or more of the fasteners **132c** could comprise a head that mates with a detent (not shown) positioned below the base **110**, to lock the weight bar **130c** in place against the bottom surface of the base **110**. Such detent coupling mechanisms could be formed in any suitable surface, such as one or more of the supports **107**. The weight bar **130c** could therefore slide forward along the sliding rail **137** to mate with the lower surface of the base **110** and be locked in place when not in use, or when the exercise device is folded into a storage position.

(41) In some embodiments, where a user **200** may wish to utilize the exercise device **100c** as a standing leg exercise device by holding the left and right handles **121** and **122**, respectively, the user **200** may slide the weight bar **130c** as far as possible towards the roller **102** such that the u-shaped handle of the weight bar **130c** rests underneath the roller **101**. In such embodiments, the feet **109** are preferably configured to position the underside of the continuous belt **105** a greater height than the thickness of the weight bar **130c**, such as more than 3 inches, 4 inches, 5 inches, or even 6 inches. By allowing the weight bar **130c** to slide up towards the roller **102** along the sliding rail **137**, a user **200** could easily position the weight bar **130c** out of the way by loosening fastening mechanism **136**, sliding the fastening mechanism **136** towards the roller **102** while keeping the u-shaped handle of the weight bar **130c** low and tucking it under the underside of the continuous belt **105**, and then by tightening the fastening mechanism **136**. After such a change in configuration, the user **200** could then stand on the continuous belt **105** and hold onto the left and right handles **121** and **122**, respectively, while moving the continuous belt **105** with their feet in the manners described herein.

(42) In some embodiments, a torque multiplier/reduction gear could be coupled to one or both of roller **101** and/or **102** to increase or decrease a resistance to rotation of the belt **105** as it rotates about the rollers **101** and **102**. Such a gear could comprise any suitable shape, such as the fastener **126** and **127**, which are configured to increase/decrease a resistance applied to the roller **102** to adjust a resistance applied against rotation of the belt **105** about the rollers. In some embodiments, a surface of one or both of the rollers is not smooth, to increase a friction force between an outer surface of the roller and the inner surface of the belt **105**. For example, an outer surface of the roller **102** could be configured to have 1 mm deep projections or recesses every 2 mm to increase a friction force, or could comprise a high-friction material such as carpeting or rubber. In some embodiments, the friction force of the torque gear could comprise a plurality of discrete settings, each one applying a different amount of force between the surface of the roller and the belt **105**. In one embodiment, rotating the torque gear **126** could tighten or loosen a screw against an axle upon which the roller **102** rotates, tightening or releasing, respectively, a brake pad applied to the axle to increase resistance of the **102** against a rotating motion. In another embodiment, the torque gear **126** could move a distance of the roller **102** from the roller **101**, where rotating the torque gear in **126** in one direction moves the roller **102** away from the roller **101**, tightening the belt **105** and increasing a friction force, whereas rotating the torque gear **126** in an opposite direction moves the roller **102** towards the roller **101**, loosening the belt **105** and decreasing the friction force.

(43) Exercise machines, such as the exercise machines **100**, **100a**, **100b**, and **100c** shown in FIGS. 1-7 could be utilized by users in various advantageous ways. By rotatably coupling the base **110**, the pair of handles **120**, and the weight bar **130** to one another, a user could fold the exercise machine **100** between the storage configuration shown in FIG. 1 and the ready-to-use position

shown in FIG. 3 via the intermediary position shown in FIG. 2. For example, when folding the exercise machine **100** from the ready-to-use position shown in FIG. 3 to the storage configuration shown in FIG. 1, the threaded fasteners **126** and **127** could be loosened to allow the base **110** to rotate relative to the pair of handles **120**. The user could then pull up the base **110** towards a vertical configuration substantially parallel to the pair of handles **120**, such as the vertical configuration shown in FIG. 1. Once the base **110** is in such a vertical configuration, the user could tighten the threaded fasteners **126** and **127** to hold the rotational position of the base **110** and the pair of handles **120** in place relative to one another. While the base **110** is being lifted to the vertical configuration, the user could also maneuver the weight bar **130** to rotate towards the bottom of the base **110** to rest on the bottom surface of the base **110** as shown in FIG. 1.

(44) When folding the exercise machine **100** from the storage configuration shown in FIG. 1 to the ready-to-use position shown in FIG. 3, a user could, again, loosen the threaded fasteners **126** and **127** to allow the base **110** to rotate relative to the pair of handles **120**. The user could then lift the weight bar **130** and rotate it towards the top surface of the base **110** as shown in FIG. 2 while simultaneously pulling the base **110** down such that the edge of the base **110** with the roller **101** rests on the ground, as shown in FIG. 3. Once the exercise machine **100** is in the ready-to-use position as shown in FIG. 3, the user could choose to tighten the threaded fasteners **126** and **127** to hold the base **110** and the pair of handles **120** in place relative to one another, or could leave the threaded fasteners **126** and **127** untightened depending upon need.

(45) For example, in embodiments where the tighteners **126** and **127** act to adjust a resistance acting against rotation of the continuous belt **105**, the user could tighten either tightener **126** or tightener **127** to increase resistance when the continuous belt **105** rotates about the rollers **101** and **102**. In some embodiments, the tighteners **126** and **127** could have visual indicators (not shown) that indicate how much braking force is being applied to the roller **102**, such as a “1” for 1 lb. of braking force, a “2” for 2 lb. of braking force, and a “3” for 3 lb. of braking force. Once an appropriate amount of braking force has been applied to the roller **102**, a user **200** could use the exercise device **100** to exercise their buttock muscles. In other embodiments, resistance applied against the rotation of the continuous belt **105** could be utilized by a second resistance device, such as a gear multiplier and/or a tension brake that is directly coupled to either of the rollers **101** and **102**.

(46) In one embodiment, a user **200** uses the exercise device **100** by lying down in front of the exercise device **100** with their feet on the continuous belt **105** as shown in FIG. 3. The user **200** braces their back **240** against a surface, such as resting surface **300**, such that the user remains in place relative to the exercise device **100**, even while moving their feet against the continuous belt **105**. Contemplated resting surfaces include a floor of a room, a carpet, a rug, or an exercise mat. Such resting surfaces **300** preferably have a coefficient of friction greater than 1 with materials covering the back **240**, such as cloth or human skin. Once the user's back **240** is properly braced against a surface, such as resting surface **300** or back support **150**, the user can use the exercise device **100** by moving their feet **230** against the surface of the continuous belt **105**. While FIGS. 3-7 show the user **200** performing exercises by first placing the weight pad **133** of the weight bar **130** on the user's torso, in some embodiments the user **200** may not use the weight bar **130** at all, for example by folding the weight bar **130** out of the way, by completely removing the weight bar **130** by removing the attachment mechanisms **136**, or by sliding the weight bar **130** along the base **110** until a portion of the weight bar **130** is tucked under the base **110**.

(47) In one embodiment, the user **200** “walks” the continuous belt **105** forward, in a clockwise direction from the perspective shown in FIG. 4. To do so, the user **200** could place the balls of both of their feet **230** on the top surface of the continuous belt **105** with both knees bent. The user **200** could retract their left knee while directing more of their body weight on their right foot and pushing down with their right knee to push the ball of their right foot down against the top surface of the continuous belt **105**, away from the body of the user **200**. This causes the user **200**'s right

foot to push the top surface of the continuous belt **105** towards the roller **102** and slide forward underneath the user **200**'s left foot. The user **200** then repeats this process with the opposite feet. In other words, the user **200** then retracts their right knee and directs more of their body weight on their still-retracted left foot and pushes down with their left knee to push the ball of their left foot down against the top surface of the continuous belt **105**, away from the body of the user **200**. This causes the user **200**'s left foot to push the top surface of the continuous belt **105** towards the roller **102** and slide forward underneath the user **200**'s right foot. By repeating this process, the user **200** “walks” the continuous belt **105** forward, in a clockwise direction as seen in FIG. 4.

(48) In another embodiment, the user “walks” the continuous belt **105** backwards, in a counterclockwise direction from the perspective shown in FIG. 4. To do so, the user **200** could place the balls of both of their feet **230** on the top surface of the continuous belt **105** with both knees bent. The user **200** could then extend their left knee while directing more of their body weight on their right foot and retracting their right knee to pull the ball of their right foot **230** towards the buttocks **220** of the user **200** while the ball of their right foot **230** is pressed against the top surface of the continuous belt **105**. This causes the user **200**'s right foot to pull the top surface of the continuous belt **105** towards the roller **101** while the user **200**'s left foot extends forwards. The user **200** then repeats this process with the opposite feet. In other words, the user **200** extends their right knee while directing more of their body weight on their extended left foot and retracts their left knee to pull the ball of their left foot **230** towards the buttocks **220** of the user **200** while the ball of their left foot **230** is pressed against the top surface of the continuous belt **105**. This causes user **200**'s left foot to pull the top surface of the continuous belt **105** towards the roller **101** and slide backwards underneath the user **200**'s right foot. By repeating this process, the user **200** “walks” the continuous belt **105** backwards, in a counterclockwise direction as seen in FIG. 4.

(49) In another embodiment, the user **200** uses one foot to “push” the continuous belt **105** forwards while bracing their other foot against the resting surface **300**. To do so, the user **200** could place their left foot on the resting surface **300** just next to the roller **101**. The user **200** could then place the ball of their right foot on the top surface of the continuous belt **105** and push down with their right knee to push the ball of their right foot down against the top surface of the continuous belt **105**, away from the body of the user **200**. This causes the user **200**'s right foot to push the top surface of the continuous belt **105** towards the roller **102**. The user **200** could then retract their right knee to place the ball of their right foot on the top surface of the continuous belt closer to the roller **101** and repeat this process, repeatedly pushing the top surface of the continuous belt **105** towards the roller **102** with the same right foot, rotating the continuous belt clockwise as seen in the perspective of FIG. 4. The user **200** could then switch the feet after a threshold number of sets have been performed, such as 5, 10, or 20. The user **200** could then place their right foot on the resting surface **300** next to the roller **101** and place the ball of their left foot on the top surface of the continuous belt **105**. The user **200** could then push down with their left knee to push the ball of their left foot down against the top surface of the continuous belt **105** away from the body of the user **200**. The user **200** could then retract their left knee to place the ball of their left foot on the top surface of the continuous belt closer to the roller **101** and repeat this process, repeatedly pushing the top surface of the continuous belt **105** towards the roller **102** with the same left foot, rotating the continuous belt clockwise as seen in the perspective of FIG. 4.

(50) In another embodiment, the user **200** uses one foot to “pull” the continuous belt **105** backwards while bracing their left foot against the resting surface **300**. To do so, the user **200** could place their left foot on the resting surface **300** just next to the roller **101**. The user **200** could then extend their leg and place the ball of their right foot on the top surface of the continuous belt **105** closer to the roller **102**, and retract their right knee while pushing their right foot downwards against the top surface of the continuous belt **105**, pulling it towards the body of the user **200**. This causes the user **200**'s right foot to pull the top surface of the continuous belt **105** towards the roller **101**. The user **200** could then extend their right knee to place the ball of their right foot on the top

surface of the continuous belt closer to the roller **102** and repeat this process, repeatedly pulling the top surface of the continuous belt **105** towards the roller **101** with the same right foot, rotating the continuous belt counterclockwise as seen in the perspective of FIG. **4**. The user **200** could then switch the feet after a threshold number of sets have been performed, such as 5, 10, or 20. The user **200** could then place their right foot on the resting surface **300** next to the roller **101** and extend their left leg to place the ball of their left foot against the top surface of the continuous belt **105**. The user **200** could then retract their left knee while pushing the ball of their left foot down against the top surface of the continuous belt **105** to pull the top surface of the continuous belt **105** towards from the body of the user **200**. The user **200** could then extend their left knee to place the ball of their left foot on the top surface of the continuous belt closer to the roller **102** and repeat this process, repeatedly pulling the top surface of the continuous belt **105** towards the roller **101** with the same left foot, rotating the continuous belt counterclockwise as seen in the perspective of FIG. **4**.

(51) The user **200** could increase the difficulty of the exercise by lifting their buttocks **220** such that they maintain a minimum threshold distance **302** between their buttocks **220** and the resting surface **300**, such as at least 1 in., at least 4 in., at least 6 in., at least 10 in., or at least 12 in., during the exercise. By maintaining such a minimum threshold distance **302** while performing an exercise, the user's abdominal muscles and buttock muscles remain engaged without rest while performing the exercise.

(52) The user **200** could further increase the difficulty of the exercise by resting the weight bar **130** on their torso **210** while maintaining the minimum threshold distance **302**. While only one weight is shown on the weight post **131**, additional weight or less weight could be applied to the weight bar **130** by coupling a suitable number of weights **135** to the weight post **131**. As previously stated, in some embodiments, the user **200** might not want to use the weight bar **130** during an exercise, in which case the weight bar **130** could be folded back against the top or underside of the base **110**, the weight bar **130** could be removed from the base **110** altogether by removing fasteners **136**, or the weight bar **130** could be slid underneath the base **110** using a slider, such as the sliding rail **137**.

(53) When using the weight bar **130**, the user **200** generally first couples one or more weights to the weight post **131** of the weight bar **130**. Then, the user **200** could position themselves under the weight bar **130** such that the pad **133** rests on a portion of the torso **210** of the user **200**, such as the user's chest, abdominals, or pelvic area. The user **200** could then brace their back **240** against the resting surface **300** and lift their buttocks **220** such that they maintain a minimum threshold distance **302** between their buttocks **220** and the resting surface **300**. By lifting their buttocks **220** the minimum threshold distance **302**, the user **220** also raises the weight bar **130** via the pad **133**, thereby engaging their buttock and core muscles while performing the exercise. Any of the aforementioned exercises could then be performed while the user maintains the minimum threshold distance **302** between their buttocks **220** and the resting surface **300**.

(54) When a user **200** uses the exercise machine **100a** with the weight bar **130a** shown in FIG. **5**, the user **200** could use the weight bar **130a** in a similar manner as the weight bar **130**. The user **200** could either fold the weight bar **130a** out of the way or remove the weight bar **130a** completely by removing the attachment mechanisms **136** if the user **200** does not wish to use the weight bar **130a**. In embodiments with a slider, the weight bar **130a** could be slid along the length of the base **110** to move it out of the way. The user **200** could also add or remove additional weights to the weight posts **131a** and **133a** to increase resistance. Once the user **200** positions themselves below the weight bar **130a**, the user **200** could push the weight bar **130a** upwards with their torso **210** as shown to exercise in a similar manner to exercising with the weight bar **130** of the exercise device **100**. When the user **200** wishes to take a break, the user **200** could then rest their buttocks **220** on the resting surface **300**, which would then cause the two weight supports **141** and **142** to drop down to the resting surface **300**. Such a position allows the weight of the weight bar **130a** to be fully supported by the two weight supports **141** and **142**, and not be supported by the torso **210** of the

user **200**, allowing the user **200** to rest in between sets. When the user **200** wishes to, again, exercise, the user **200** could then push the weight bar **130a** upwards with their torso **210** as shown to continue the exercise. In some embodiments, when a user **200** does not wish to use the two weight supports **141** and **142**, the user **200** could rotate the weight supports **141** and **142** to be substantially parallel to the side bars of the weight bar **130a** and tighten the attachment mechanisms **144** to prevent the weight supports **141** and **142** from rotating relative to the body of the weight bar **130a**.

(55) When a user **200** uses the exercise device **100b** with the weight bar **130b** and the back support **150** shown in FIG. 6, the user **200** could use the weight bar **130b** in a similar manner as the weight bar **130**. The user **200** could either fold the weight bar **130b** out of the way or remove the weight bar **130b** completely by removing the attachment mechanisms **136** if the user **200** does not wish to use the weight bar **130b**. In embodiments with a slider, the weight bar **130b** could be slid along the length of the base **110** to move it out of the way. Similarly, the user **200** could either fold the back support **150** out of the way or remove the back support **150** completely by removing the attachment mechanisms **136** if the user **200** does not wish to use the back support **150**. When a user **200** wishes to use both the weight bar **130b** and the back support **150**, the user **200** could first pull up on the back support **150** to ensure the back support **150** rests on a resting surface while unfolding the exercise device **100b** to the ready-to-use position. The user **200** could then couple a suitable number of weights **135** to the weight post **131** and holds the weights **135** in place using the clamp **170**. The user **200** could then position the weight bar **130b** on their torso **210** while bracing their back **240** against the back support **150**. While the back support **150** is shown here as coupled to the exercise device **100b** via the attachment mechanism **136**, detached back supports could be used, such as a yoga mat, which could be coupled to the exercise device **100b** by placing the mat underneath the feet **109** of the exercise device **100b** to hold the detached back support in place relative to the exercise device **100b**.

(56) When a user **200** uses the exercise device **100c** with the weight bar **130c** shown in FIG. 7, the user could place the weight bar **130c** on the resting surface **300** such that the support posts **141** and **142** hold the weight of the weight bar **130c** when the exercise device **100c** is in the ready-to-use position shown. The user could then couple any number of weights, such as the weights **137c** and **139c** to the weight posts **136c** and **138c**, respectively. In some embodiments, the user **200** could use clamps, such as clamp **170** in FIG. 6, to hold the weights in place on the weight posts **136c** and **138c**. Since the support posts **141** and **142** hold the weight bar **130c** above the resting surface **300** in the ready-to-use position, the user **200** does not need to lift the weight bar **130c** from the resting surface **300** to place the weights on the weight posts **136c** and **138c**. The user **200** could then position themselves underneath the weight bar **130c** as shown in FIG. 7, and lift the weight bar **130c** upwards using their torso **210** to press upwards on the padding **133c**, thereby maintaining a minimum distance between their buttocks **220** and the resting surface **300**. The user **200** could maintain that minimum distance **302** while moving the continuous belt **105** with their feet **230** in any of the manners described above. When the user **200** wishes to rest, the user **200** could rest their buttocks **220** on the resting surface **300**, allowing the support posts **141** and **142** to hold the weight of the weight bar **130c** between sets.

(57) In some embodiments, the support posts **141** and **142** could be configured to fold out of the way to allow the user **200** to rest the weight bar **130c** directly on the user's torso **210** while the user's back **240** is resting on a resting surface. Such a configuration allows the user **200** to perform hip thrusts while the user's back **240** is resting on the resting surface, but also allows the user **200** to allow the support posts **141** and **142** to be used to prevent the weight bar **130c** from resting directly on the user's torso **210** in other configurations. Enabling such flexibility could be appropriate, for example, when the weight bar **130c** is resting on a different portion of the user's torso **210** or when the weight bar **130c** is holding a different amount of weight. The support posts **141** and **142** could be biased to rest in a disengaged position substantially parallel to the weight bar **130c** or in an

engaged position substantially non-parallel to the weight bar **130c** (e.g. perpendicular, acute, or obtuse), similar to a bicycle kickstand.

(58) In some embodiments, the user **200** could even use an exercise device, such as exercise device **100**, while standing to exercise their leg and gluteal muscles. For example, the user **200** could stand on the top surface of the continuous belt **105** while holding onto the handles **120**. While the user grips the left handle **121** with their left hand and the right handle **122** with their right hand, respectively, the user **200** could then use their feet **230** to move the top surface of the conveyor belt **105** in either direction. For example, the user **200** could brace one foot against the top surface of the conveyor belt **105** while extending their other leg to pushing the top surface of the conveyor belt **105** towards the roller **101** using the ball of their foot, and could repeat the process with opposite feet to rotate the conveyor belt **105** about the rollers **101** and **102**. Alternatively, the user **200** could brace one foot against the top surface of the conveyor belt **105** while pushing the conveyor belt **105** towards roller **102** using the heel of their foot while pulling on the handles **120** with their arms, and repeat the process with opposite feet to rotate the conveyor belt **105** about the rollers **101** and **102**. Such a configuration allows the user **200** to use the exercise device **100** in multiple ways to exercise the user's leg and buttock muscles.

(59) As discussed elsewhere herein, the weight bar **130** applies a downward force on the user's pelvis area/torso area. However, it is also contemplated that the downward force may be applied with a resistance band **805** as shown and described in relation to FIGS. **8A-D** and **9A-C**. More particularly, as shown in FIG. **8A**, another embodiment of the exercise device **100d** that is more compact is shown. The exercise device **100d** may have a base **110** with a continuous belt system having a continuous belt **105** that is taut between at least two rollers **101**, **102**, as described elsewhere herein. A back support **150** may be attached in front of the continuous belt system of the base **110**. By way of example and not limitation, the base **110** having the continuous belt system may be elevated height-wise relative to the back support **150**. By way of example and not limitation, the base **110** may have the continuous belt **105** inclining downwards towards the back support **150** such that the incline adds a further level of challenge for the user when using the exercise device **100d**. The back support **150** may be where a user **200** may rest his back **240** on (see FIG. **8C**) when using the exercise device **100d**, and one or more resistance bands **805** may be attached to the outer frame **806** of the back support **150** to provide a downward resistive force on the user's torso and pelvis area **210-220**, especially when the user lifts his pelvis/buttocks **220** off the back support **150** for a more intense exercise session. By way of example and not limitation, the outer frame **806** may surround the lying surface **151** of the back support **150**, and the outer frame **806** may have adjustable gripping handles **804** and attachment rings **802** on its left and right sides of the frame.

(60) The gripping handles **804** may be used by the user **200** (see FIG. **8C**) during the exercise routines, described elsewhere herein, to help lift his pelvis and buttocks **220** off the ground while lying on the back support **150**. The user **200** may also use the gripping handles **804** to obtain better leverage when moving his legs forwards or backwards along the continuous belt **105** of the base **110** during the exercise routines, described elsewhere herein. By way of example and not limitation, the attachment rings **802** may replace the weight bar **130** in the other embodiments, where one or more resistance bands **805** are attached to the attachment rings **802** and stretched laterally along the outer frame **806** to provide the resistance force on the torso and pelvis area **210-220** of the user **200**, which the weight bar **130** would have provided in other embodiments. Alternatively, the attachment rings **802** may be used in conjunction with the different examples of weight bars **130** in the other embodiments of the exercise device **100** to add more resistive force on the user's torso and pelvis area **210-220**. By way of example and not limitation, the gripping handles **804** and the attachment rings **802** may be adjustable and slidable along the length of the outer frame **806** to accommodate the usage of the exercise device by users of different sizes (e.g., users having different heights).

(61) The outer frame **806** may be designed to hold a lying surface **151** therebetween for the user **200** (see FIG. **8C**) to rest his back **240** and buttocks/pelvis **220** (at least initially) on such surface. By way of example and not limitation, the outer frame **806** may have a rectangular, trapezoidal, oval, or any other shape that is big enough to surround the user **200** while he is lying on the lying surface **151** of the back support **150**. By way of example and not limitation, the outer frame **806** may have two longitudinal sides **808** and at least one lateral side. By way of example and not limitation, the front edges of the outer frame **806** of the back support **150**, which such front edges may extend from the two longitudinal sides **808**, may be affixed and attached to the trapezoidal frame of the base **110**. Alternatively, the front edges of the outer frame **806** of the back support **150** may be rotatably attached to the trapezoidal frame of the base **110** using the fasteners **136** (see FIG. **6**), as described elsewhere herein.

(62) By way of example and not limitation, each of the longitudinal sides **808** of the outer frame **806** may have a plurality of slots **810a-b** for the slidable insertion and adjustment of the gripping handles **804** and attachment rings **802**. The slots **810a-b** may extend along the length of the left and right longitudinal sides **808** of the outer frame **806**, either extending fully or partially along the length. By way of example and not limitation, the slots **810a-b** may extend partially along the center length portion of the longitudinal sides **808**. By way of example and not limitation, the slots **810b** for the gripping handles **804** may be on upper surfaces (see FIG. **8D**) of the longitudinal sides **808** of the outer frame **806**, and the slots **810a** for the attachment rings **802** may be on inner-side surfaces of the longitudinal sides **808** of the frame **806**. Alternatively, the upper slots **810b** may be used for attachment rings **802** and outer or inner-side slots **810a** for the gripping handles **804**. By way of example and not limitation, the gripping handles **804** and/or attachment rings **802** may be inserted within the slots **810a-b** using slot openings **812** (see FIG. **8A**) on the front and rear tips of the longitudinal sides **808** of the outer frame **806**. By way of example and not limitation, the slots **810a-b** may be T-slots or V-slots, or a combination thereof. By way of example and not limitation, the longitudinal sides **808**, and the outer frame **806** in general, may be made from an aluminum or steel alloy. As shown in FIG. **8D**, and by way of example and not limitation, the outer frame **806** and particularly its longitudinal sides **808** may be made from extrusion profile frames having T-slots or V-slots and being made of aluminum or steel. The extrusion profile frames may be made from 80/20®. By way of example and not limitation, the outer frame **806** and its longitudinal sides **808** may be made from a rigid polymer or composite material instead of aluminum or steel alloy.

(63) By way of example and not limitation, the attachment rings **802** may be slidably adjustable along the length of the outer frame **806** using the inner-side slots **810a**, as described elsewhere herein. As shown in FIG. **8B**, the attachment rings **802** may be adjusted to a desired position on the longitudinal sides **808** of the outer frame **806**. By way of example and not limitation, the attachment rings **802** may be fixed into place when slid to the desired position along the length of the outer frame **806**. By way of example and not limitation, there may exist at least one attachment ring **802** on each longitudinal side **808** of the outer frame **806**. As shown in FIG. **8A**, each longitudinal side **808** of the outer frame **806** may have a plurality of attachment rings **802** (e.g., two attachment rings) slidable on the inner-side slots **810a**. The plurality of attachment rings **802** on each longitudinal side **808** may allow for the binding of multiple resistance bands.

(64) As shown in FIG. **9A**, and by way of example and not limitation, the attachment ring **802** may have a body **906** with a ring portion **907** on one end and a sliding portion **905a** on the other end, where the sliding portion **905a** is inserted in a through hole **904a** of an attachment block **902a** and extends on the other side of the attachment block **902a**. By way of example and not limitation, the attachment block **902a** may have a plurality of through holes **904a** for inserting the sliding portion **905a** of the body **906** of the attachment ring **802** to adjust the ring portion **907** along the length of the body of the attachment block **902a**. By way of example and not limitation, the ring portion **907** may be a hook (e.g., a snap/carabiner hook) instead of a ring, or any other attachment/fastening mechanism that can be considered an attachment component similar to the attachment ring **802**.

Such attachment components, similar to the attachment ring **802**, may also be slidably adjusted along the length of the outer frame **806** using slots **810a**, as described elsewhere herein and as shown in FIG. **8B**. By way of example and not limitation, the sliding portion **905a** of the body **906** of the attachment ring **802** may be a shaft. By way of example and not limitation, the sliding portion **905a** may be a threaded shaft and the attachment block holes **904a** may be correspondingly threaded to fasten the threaded shaft that is the sliding portion **905a** to the attachment block **904a**. In another example, the attachment ring **802**, or an attachment component in general, may have one unitarily formed body with the ring portion **907** extending on one end of the attachment block **902a** and the sliding portion **905a** extending on the opposite end of the attachment block **902a**. By way of example and not limitation, the attachment ring **802** may be made from a metal alloy, such as aluminum or steel, or be made from a rigid polymer material or composite material.

(65) By way of example and not limitation, the gripping handles **804** may be slidably adjustable along the length of the outer frame **806** using the upper slots **810b**, as described elsewhere herein. As shown in FIG. **8B**, the gripping handles **804** may be adjusted to a desired position on the longitudinal sides **808** of the outer frame **806**. By way of example and not limitation, the gripping handles **804** may then be fixed into place when slid to the desired position along the length of the outer frame **806**. By way of example and not limitation, there may exist one gripping handle **804** on each longitudinal side **808** of the outer frame **806**.

(66) As shown in FIG. **9B**, and by way of example and not limitation, the gripping handle **804** may have a shaft with a gripping portion **910** on one end and a sliding portion **905b** on the other end, where the sliding portion **905b** is inserted through a through hole **904b** of a handle block **902b** and extends on the other side of the handle block **902b**. By way of example and not limitation, the handle block **902b** may have a plurality of through holes **904b** for inserting the sliding portion **905b** of the shaft of the gripping handle **804** to adjust the gripping portion **910** along the length of the body of the handle block **902b**. By way of example and not limitation, the gripping portion **910** may be thicker than the sliding portion **905b** of the gripping handle **804**. By way of example and not limitation, the gripping portion **910** may be cylindrical shaped or prism shaped, where the sides of the lateral faces of the prism act as gripping surfaces. By way of example and not limitation, the outer end of the gripping portion **910** may have a dome shape **908** for ease of gripping the gripping handle **804**. By way of example and not limitation, the sliding portion **905b** may be a threaded shaft and the handle block holes **904b** may be correspondingly threaded to fasten to the sliding portion **905b**. By way of example and not limitation, the sliding portion **905b** may also be a separate component than the gripping portion **910**, and the sliding portion **905b** may be a threaded shaft to screw in a threaded bore on the opposite side of the gripping portion **910** of where the dome shape **908** is located. In another example, the gripping handle **804** may have one unitarily formed body with the gripping portion **910** extending on one end of the handle block **902b** and the sliding portion **905b** extending on the opposite end of the handle block **902b**. By way of example and not limitation, the gripping handle **804** may be made from a metal alloy, such as aluminum or steel, or be made from a rigid polymer material or composite material.

(67) As shown in FIG. **8C**, one or more resistance bands **805** may be stretched laterally between the longitudinal sides **808** of the outer frame **806** and attached to the attachment rings **802** such that the resistance band **805** biases again the torso **210** of the user **200**. The usage of the resistance band **805** with the attachment rings **802** may be an alternative method to provide the same function as the weight bar **130**, shown in other figures and described elsewhere herein, such as providing an added resistance on the torso **210** and pelvis/buttocks region **220** of the user **200** during using the exercise device **100d** when the person lifts his pelvis and buttocks off from the ground and the back support **150**. The usage of the one or more resistance bands **805** with the attachment rings **802**, instead of the weight bar **130**, may make the assembly and operation of the exercise device **100d** easier since less mechanical parts need to be combined and used during the workout. Alternatively, the one or more resistance bands **805** may be used in conjunction with the weight bar **130** of the

other embodiments of the exercise device. By way of example and not limitation, the one or more resistance bands **805** may each come in different resistance forces ranging from 10 to 100 pounds. As shown in FIG. **9C**, each resistance band **805** may have an elastic body **912** with binding mechanisms **914** at the ends of the elastic body **912**. By way of example and not limitation, the elastic body **912** may be rope-shaped having an elastically cylindrical body or an elastically rectangular body. By way of example and not limitation, the binding mechanisms **914** may be snap/carabiner hooks, where each binding mechanism **914** may be on one end of the rope-shaped body of the elastic body **912**.

(68) It will be appreciated from the foregoing that the exercise device systems and methods disclosed herein can be adapted to a wide variety of uses systems, and that systems employing this system and method can be operated to exercise gluteal muscles as will be suitable to different applications and circumstances. It will therefore be readily understood that the specific embodiments and aspects of this disclosure described herein are exemplary only and not limiting, and that a number of variations and modifications will suggest themselves to those skilled in the pertinent arts without departing from the spirit and scope of the disclosure.

Claims

1. An exercise device for strengthening gluteus muscles and hamstrings, the exercise device comprising: a base having a belt system with a continuous belt that is taut between at least two rollers; a back support attached to a front of the base, the back support having a lying surface between an outer frame of the back support, the outer frame having a first longitudinal side and a second longitudinal side; a first attachment component coupled to the first longitudinal side of the outer frame; and a second attachment component coupled to the second longitudinal side of the outer frame; a resistance band attached to the first and second attachment components to apply a downward force on a torso of a user when the user is lying on the lying surface of the back support.
2. The exercise device of claim 1, further comprising a first gripping handle slidably coupled to a first lengthwise slot of the first longitudinal side, and a second gripping handle slidably coupled to a second lengthwise slot of the second longitudinal side.
3. The exercise device of claim 2, wherein the first lengthwise slot is located on a first inner surface of the first longitudinal side and the second lengthwise slot is located on a second inner surface of the second longitudinal side.
4. The exercise device of claim 3, wherein a third lengthwise slot is located on a first upper surface of the first longitudinal side and a fourth lengthwise slot is located on a second upper surface of the second longitudinal side.
5. The exercise device of claim 2, wherein the first gripping handle comprises a shaft body with a gripping portion on one end of the shaft body and a sliding shaft on another end of the shaft body, the sliding shaft inserted through an handle block of the first gripping handle.
6. The exercise device of claim 1, wherein the first longitudinal side has a first lengthwise slot and the second longitudinal side has a second lengthwise slot, the first attachment component being slidably coupled to the first lengthwise slot of the first longitudinal side of the outer frame, and the second attachment component slidably coupled to the second lengthwise slot of the second longitudinal side of the outer frame.
7. The exercise device of claim 6, wherein the first and second lengthwise slots are either V-slots or T-slots.
8. The exercise device of claim 1, wherein the first attachment component comprises a first body having a first attachment ring on one end of the first body and a first sliding shaft on another end of the first body, the first sliding shaft inserted through an attachment block of the first attachment component.
9. The exercise device of claim 1, wherein the first and second attachment components each

comprise an attachment ring.

10. The exercise device of claim 1, wherein the first and second attachment components each comprise a snap hook.

11. An exercise device for strengthening gluteus muscles and hamstrings, the exercise device comprising: a base having a belt system with a continuous belt that is taut between at least two rollers, the belt system being at an incline; a back support attached to a front of the base, the back support having a lying surface between an outer frame of the back support, the outer frame having a first longitudinal side and a second longitudinal side, the first longitudinal side having a first lengthwise slot and the second longitudinal side having a second lengthwise slot; a first attachment component slidably coupled to the first lengthwise slot of the first longitudinal side of the outer frame; and a second attachment component slidably coupled to the second lengthwise slot of the second longitudinal side of the outer frame; wherein the first and second attachment components are configured to have a resistance band attached thereto for the resistance band to apply a downward force on a torso of a user when the user is lying on the lying surface of the back support; wherein the base is elevated height-wise relative to the back support and the belt system inclines downwards towards the back support.

12. The exercise device of claim 11, further comprising a first gripping handle slidably coupled to a third lengthwise slot of the first longitudinal side, and a second gripping handle slidably coupled to a fourth lengthwise slot of the second longitudinal side.

13. The exercise device of claim 12, wherein the first lengthwise slot is located on a first inner surface of the first longitudinal side and the second lengthwise slot is located on a second inner surface of the second longitudinal side.

14. The exercise device of claim 13, wherein the third lengthwise slot is located on a first upper surface of the first longitudinal side and the fourth lengthwise slot is located on a second upper surface of the second longitudinal side.

15. The exercise device of claim 12, wherein the first gripping handle comprises a shaft body with a gripping portion on one end of the shaft body and a sliding shaft on another end of the shaft body, the sliding shaft inserted through an handle block of the first gripping handle.

16. The exercise device of claim 11, wherein the first and second longitudinal sides are made from extrusion profile frames.

17. The exercise device of claim 16, wherein the first and second lengthwise slots are either V-slots or T-slots.

18. The exercise device of claim 11, wherein the first attachment component comprises a first body having a first attachment ring on one end of the first body and a first sliding shaft on another end of the first body, the first sliding shaft inserted through an attachment block of the first attachment component.

19. The exercise device of claim 11, wherein the first and second attachment components each comprise an attachment ring.

20. The exercise device of claim 11, wherein the first and second attachment components each comprise a snap hook.
